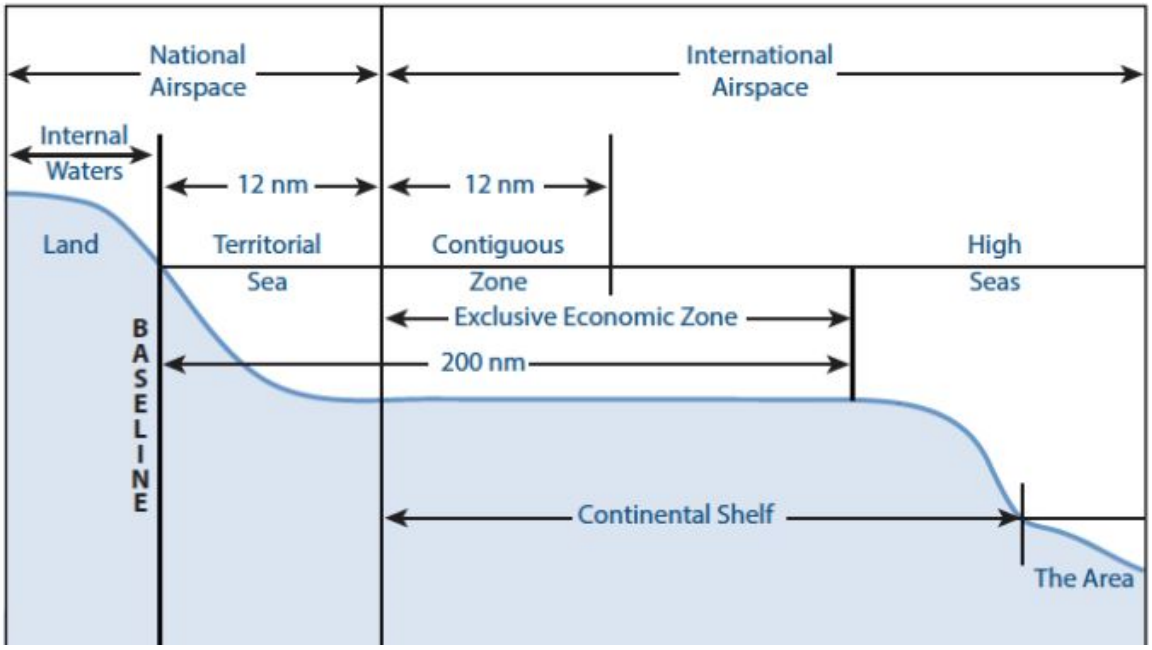
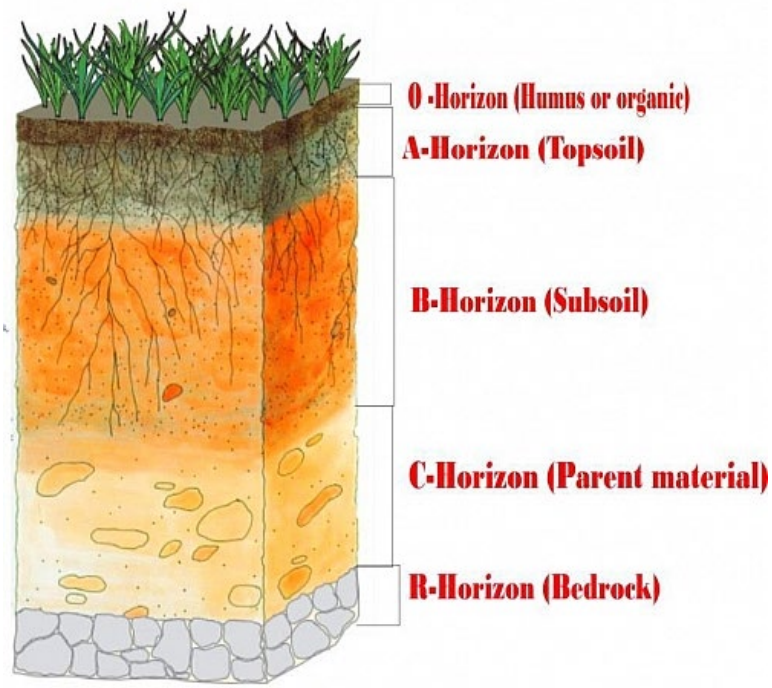
- 

- Baseline
- It is the average line where the land ends and the ocean begins.
- Internal Waters
- It includes all the water bodies along the landward side of the baseline.
- Territorial Sea
- It is 12 nm from the baseline.
- A country exercises sovereign jurisdiction on the bottom, surface, and air. However, innocent passage is allowed with restrictions.
- Contiguous Zone
- It is up to 24 nm from the baseline.
- Sovereign jurisdiction over the bottom and surface but not in the air.
- Exclusive Economic Zone
- up to 200 nm from the baseline.
- A country can exploit the resources on the ocean floor.
- It extends beyond 200 nm if the continental shelf crosses it.
- It will be allowed till the edge of the continental shelf.
- High seas
- International waters come under International Seabed Authority.

Biogeography (7:12 PM)

- It is the study of the soil in relation to vegetation.
- Soil
- The loose surface material consists of inorganic and organic matter and is the source of water and nutrients for the vegetation.
- True soil is characterized by distinct horizons and the capacity to support vegetation.
- Horizon
- Horizon is a distinguishable layer in the soil that has certain chemicals and a distinct colour.
- Regolith
- The layer of unconsolidated materials is derived from weathering of the parent rock.

- Soil Profile
- It is the vertical arrangement of the horizons down to the parent rock material.
- 

Soil Formation (7:45 PM)

- Soil depends on five major factors: Parent rock, climate, organism, relief, and time.
- Parent Rock Material
- It decides the texture, colour, and basic mineral composition of the soil.
- For Example Quartzite rocks, result in a sandy texture.
- However, the role of the parent rock decreases with time.
- Climate
- It is the most dominant factor which affects the soil directly, by controlling weathering, percolation, and other processes.
- It results in the development of different horizons.
- Organisms
- Plant roots, burrowing animals, and microorganisms help in loosening up the soil and better air and nutrient circulation.
- Relief
- It determines the thickness of the soil.
- Time
- It controls the other factors and determines the maturity of soil.
- **Soil Formation Processes**
- Transformational Process
- The change of soil constituents from one form to another through breakdown, weathering, decay, etc.
- Translocational Process
- It involves the movement of soil constituents from one horizon to another within the soil profile.
- Eluviation: The downward transport process through which the minerals are removed from the top layer

- Illuviation: It is the reverse of eluviation, in which matter accumulates in the lower horizon.
- Leaching: the percolating water removes humus and the soluble bases from the upper horizon and deposits them in the lower horizon through the process of leaching.
- Leaching is more active in humid regions.

The Topic for the next class: Soil (continued), Mapping, and Indian Geography.
Geography Class 35

A brief discussion on the previous class (5:10 PM)
Laterization (5:34 PM)

- It is active in hot and wet equatorial regions.
- Heavy leaching removes silica and other minerals, except for iron and aluminium, which accumulate at the surface, forming a hard crust.
- This process is called Laterization.
- It results in laterite soil.
- It is slightly acidic.

Podzolization(5:42 PM)

- It occurs in temperate regions and mountainous regions.
- The decomposition of coniferous litter releases organic acids.
- During precipitation, the water mixing through these organic acids creates a strongly acidic solution that removes iron, aluminium, organic matter, and other minerals except silica.
- This leaves the top layer acidic.
- It results in podzol soil.

Calcification (5:55 PM)

- It occurs in regions where evaporation exceeds precipitation.
- Water rises through capillary action in the soil.
- It causes the deposition of calcium carbonate in the top layer
- It is more active in temperate grasslands.
- It results in the black earth

Salinization

- It involves the accumulation of highly soluble sodium and magnesium salts, in the topsoil.
- It is due to excessive evaporation, and very less precipitation.
- It is active in arid and semi-arid regions
- It is also common in the regions of flood irrigation.

Gleisation

- It occurs in water-logged conditions.
- The colour of topsoil changes due to the reduction reactions.
- It involves the accumulation of organic matter in the upper layers of the soil.
- It is common in coastal regions along lagoons and deltas.

Distribution of Soils (6:04 PM)

- Two types of soil- **Pedocal** and **Pedalfers**
- Pedocal
- Net upward movement of water
- evaporation exceeds precipitation
- Types are Prairie soil, chernozem soil, chestnut soil, Desert soil

- Pedalfer
- Net downward movement
- Precipitation exceeds evaporation
- Types are Laterite soil, red soil, Podzol soil
- **USDA Classification**
-

Entisol	Fresh Alluvium
Inceptisol	Old alluvial, volcanic soil
Spodosol	Podzol soil
Alfisol	Leached Soil with Al and Fe
Ultisol	Leached Soil
Oxisol	The top layer is weathered with rich Al and Fe
Vertisol	Black Soil
Histosal	Peaty Soil rich in Organic Matter
Mollisol	Prairie Soil
Aridisol	Desert Soil

Mapping (6:18 PM)

- Europe
- Seas: Mediterranean Sea, Adriatic Sea, Black sea, Caspian sea
- Mountains: Ural mountains, Alps mountain, Caucasus mountain, Mt Elbrus
- Countries: Vatican City
- Cities: Istanbul, Rome
- North America:
- Sea: Californian bay, Gulf of Mexico, Bering sea
- Mountains: Rocky Mountains, Appalachian mountains, Mt Denali
- Places: Death Valley
- Latin America: Except for Greenland, Canada, and USA, all other countries in the North and South American continent.
- South America
- Places: Drake Passage, Valdes Peninsula, Angel falls
- Mountains: Andes Mountains, Mt Aconcagua, Ojas del salado
- Rivers: Amazon, Lake Orninoco
- **Oceania (7:18 PM)**
- Countries: Australia, New Zealand
- Island groups are divided into Micronesia, Melanesia, and Polynesia
- Micronesia includes Caroline Island, Kiribati, etc
- Melanesia includes New Guinea, Fiji, etc

- Polynesia includes Hawai, New Zealand, Tonga island, Easter Island
- Places: Great Dividing Range, Great Barrier Reef, Mt Wilhelm, Southern Alps
- Antarctica
- Antarctic treaty System, 1961 held that Antarctica can be used only for peaceful experiments.
- The base stations of India in Antarctica are **Dakshin Gangotri, Maithri, and Bharati.**
- **India (7:38 PM)**
- We have both land and maritime neighbours.
- Land Neighbours are Pakistan, Afghanistan, China, Nepal, Bhutan, Bangladesh, and Myanmar.
- Afghanistan shares a border with India along **Wakhan Corridor**.
- Countries arranged in order of length of the border with India: Bangladesh(4000 km)> China(3500 km)>Pakistan(3300 km)>Myanmar(1700 km)>Nepal(1650 km)>Bhutan(700 km)>Afghanistan(100 km)
- Countries with maritime boundaries: Bangladesh, Pakistan, Myanmar, Sri Lanka, Maldives, Indonesia, Thailand
- Union territories of India: Andaman and Nicobar islands, The Government of NCT of Delhi, Ladakh, Puducherry, Lakshadweep, Jammu and Kashmir, Chandigarh, and Dadra and Nagar Haveli and Daman and Diu.
- The highest peak in India is **K2** and the Second highest peak is **Kanchenjunga**
- The lowest point in India is in **Kuttanad** (Kerala)

The topic for the next class: Physical geography of India

Geography Class 36

A brief discussion of the Previous Class (5:11 PM)

Mapping(5:25 PM)

- Sikkim and Meghalaya share a border with only one state.
- Sikkim shares a border with West Bengal only. Meghalaya Shares a border with Assam only.
- Telangana doesn't share a border with Odisha.
- Madhya Pradesh doesn't share a border with Jharkhand.
- Yamuna flows through Himachal Pradesh as well.

Physiography of India (7:53 PM)

Subcontinent

- Subcontinent is the large landmass i.e. part of the continent but is considered separate from the rest of the continent.
- For example, the Indian subcontinent, Alaskan Subcontinent, etc.
- Reasons for considering India as a subcontinent.
- (a) Geographical reasons
- Geographically, the Indian subcontinent has well-defined physical boundaries, such as the Himalayas and the Hindukush in the north, Purvanchal in the east, and the Indian Ocean in the South.
- (b) Geological reasons
- The countries of the Indian subcontinent share common geological history, such as the formation of the Himalayas, northern plains, etc.
- (c) Common climatic conditions
- All these countries experience monsoon types of climate.